

Facilities Managers

80 decision-ready facilities questions linking current owner records, correctly scoped telemetry, evidence sufficiency, uncertainty and safe failure.

STAKEHOLDER

Facilities managers coordinating asset and service information, space support, maintenance, contractors, public-facing services, incidents, lifecycle and sustainability while competent technical, security and safety owners retain authority.

QUESTION RANGE

Eighty numbered records across eight decision areas, from current operational review through long-term resilience and capital evidence.

EVIDENCE MODEL

Current owner-controlled CMMS, asset, BMS, utility, room, service, contractor, permit, security, safety and incident sources, supported only by minimum proposed privacy-preserving telemetry with verified mapping, timing, quality, provenance and permissions.

CORE PRINCIPLE

Use current authoritative evidence and only the minimum verified telemetry needed to prioritise inspection, coordination and escalation. Never operate systems, authorise work or access, certify compliance or safety, infer individual behaviour, or override competent facilities, security, health-and-safety or emergency authorities.

1. Scope and design principles

This catalogue supports evidence-aware facilities decisions. It does not transfer authority from asset owners, competent engineers, room and service owners, cleaning and waste services, security, health-and-safety, emergency command or other responsible professionals.

DECISION SUPPORT, NOT REMOTE CONTROL

The service may perform authorised lookup, observation, calculation, estimate, forecast, comparison, diagnosis and recommendation. It never operates assets, acknowledges alarms, raises permits, grants access or authorises work.

PROPOSED UNTIL INDEPENDENTLY VERIFIED

Sensors, mappings, analytics and integrations remain design requirements until deployed, commissioned, permitted and confirmed healthy for the named decision interval.

CURRENT OWNER SOURCES FIRST

CMMS, BMS, utility, room, service, permit, contractor, security, safety and incident owners govern their domains. Telemetry cannot override owner-issued status.

PRIVACY AND IDENTITY MINIMISATION

Use anonymous aggregate occupancy and numerical environmental or asset measures only where proportionate. Never identify, track or infer individual behaviour.

EXACT ROOM, ASSET AND SERVICE SCOPE

A claim requires verified room, zone, meter, asset and service-path mapping. Corridor or adjacent-room evidence cannot stand in for room-level evidence.

HONEST ANSWERABILITY AND SAFE FAILURE

Name the operation, interval, minimum evidence, uncertainty and owner. Clarify, qualify, defer or return not assessable when evidence or authority is insufficient.

2. How to read each question record

Every record states the facilities decision, exact evidence scope, named operation, uncertainty, authority boundary and safe next action.

FACILITIES MANAGERS QUESTION	A natural question a facilities manager would ask to prioritise, coordinate, compare, prepare, escalate or evidence a building-service decision.
SENSORS AND TELEMETRY	Only physically plausible, privacy-preserving proposed evidence. The field names asset or space scope, timing, aggregation, mapping, calibration or commissioning, uptime, gaps, drift and provenance limits.
AUTHORITATIVE SUPPORTING DATA	Current owner-controlled CMMS, asset, BMS, utility, room, service, contractor, permit, security, safety and incident sources, including effective, observed and retrieved times, permissions, staleness and conflicts.
ANALYSIS REQUIRED	The named operation, decision interval, minimum evidence, method, comparator or baseline, uncertainty and safe failure. Diagnosis ranks plausible explanations; it does not prove causality.
ANSWER BOUNDARY	What the service must not infer, disclose, promise, certify, authorise or control, and when it must clarify, qualify, defer or return not assessable.
WHY THIS IS USEFUL	The concrete facilities action supported: inspect, coordinate, escalate, prepare evidence, propose a window, review a lifecycle option or identify an evidence gap.

IMPLEMENTATION READINESS

R1 - OWNER-DATA READY Primarily authoritative lookup or calculation from current owner-issued records; proposed telemetry is optional context.	R2 - VERIFIED INTEGRATION Requires current cross-system integration and/or proposed telemetry with verified mapping, timing, quality and permission.	R3 - RESTRICTED OR HIGHER-CONSEQUENCE Requires restricted security, permit, safety, incident or critical-infrastructure governance and competent-authority interpretation.
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3. Worked question-to-evidence example

A morning operational-priority request is translated into source, mapping, quality and authority gates before any recommendation is given.

EXAMPLE QUESTION

Before opening today, which unresolved faults and facilities requests need my review first, and which are confirmed, suspected or still unverified?

1. RESOLVE THE DECISION

Confirm opening milestone, affected services, current owners and approved priority rules.

2. VERIFY AUTHORITATIVE DATA

Use CMMS, asset, BMS, utility, room, closure and handover owners; preserve versions and times.

3. VERIFY OPTIONAL TELEMETRY

Require exact mapping, aligned clocks, current commissioning/calibration, cadence, uptime, gaps, drift and provenance.

4. PERFORM THE RIGHT OPERATIONS

Use lookup and evidence sufficiency first, then deduplication and priority calculation; keep estimates and diagnosis separate.

5. PRESERVE AUTHORITY

Never acknowledge alarms, operate assets, issue work, grant access, certify safety or override responsible owners.

6. COMMUNICATE ANSWERABILITY

Return prioritised for review, owner-deferred, clarification needed or not assessable, with uncertainty and next owner check.

1. Asset status, service requests and operational overview

Build a current, evidence-qualified operational picture and decide what needs review, coordination or escalation.

FM-001 MORNING OPERATIONAL PRIORITIES

L2 calculation | R2 verified integration

FACILITIES MANAGERS QUESTION

Before opening today, which unresolved faults and facilities requests need my review first, and which are confirmed, suspected or still unverified?

SENSORS AND TELEMETRY

Proposed. Scope: named maintainable assets, their upstream supplies, downstream served spaces and linked requests. Evidence: run state, command/feedback, fault codes, current, temperature, pressure, flow and point-health heartbeat as relevant. Timing: source-native events plus 1-5 minute operational summaries from the last signed handover to opening. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: CMMS/service-request owner, asset-register owner, BMS or specialist-system owner and duty handover owner. Records: open request state, asset criticality, service impact, owner, priority rule, operating schedule, overrides and last signed handover. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: last signed handover to the next opening milestone. Minimum evidence: current non-duplicate requests, named owner, verified asset/service mapping and at least one current owner status source. Method: deduplicate linked requests; score consequence, duration, recurrence, service reach and evidence confidence; keep confirmed, suspected and unverified groups separate. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not turn the ranking into a work order or alarm acknowledgement; if asset/service mapping is incomplete, show a provisional queue only. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: set the first walk-round and owner-escalation order before public opening.

FM-002 COMMON-CAUSE SERVICE REQUESTS

L3 diagnosis | R2 verified integration

FACILITIES MANAGERS QUESTION

Do complaints from these rooms point to one shared plant item, utility branch or control path?

SENSORS AND TELEMETRY

Proposed. Scope: all complaint rooms and their verified HVAC, electrical, water or vertical-transport service paths. Evidence: room environmental points, terminal feedback, branch pressure/flow/current, upstream plant state and time-stamped fault events. Timing: one-minute aligned trends around each complaint plus source-native events and a matched normal period. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: CMMS owner, room/service-topology owner, BMS or specialist-system owner and maintenance-history owner. Records: complaint timestamps/categories, verified room-to-service graph, asset faults, current overrides and recent work affecting the shared path. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: diagnosis. Interval: each complaint window plus one comparable normal window. Minimum evidence: at least two independently timed complaints, verified common-path mapping and healthy points on the shared and local branches. Method: trace each room to candidate shared assets; compare event order and cross-room signatures; rank common and local explanations without claiming causality. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not merge complaints because rooms are merely nearby; show alternative explanations and mapping gaps. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: decide whether to inspect a shared upstream service or continue separate room investigations.

1. Asset status, service requests and operational overview

Build a current, evidence-qualified operational picture and decide what needs review, coordination or escalation.

FM-003 CRITICAL STATUS GAPS

L2 calculation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Which critical services lack reliable status evidence for this shift, and which gap prevents a safe priority decision?

SENSORS AND TELEMETRY

Proposed. Scope: critical utilities, lifts, plant and server-support services with authorised monitoring interfaces. Evidence: owner-issued status heartbeat, communications health, redundant state where available and last successful observation. Timing: event-driven state with owner-defined heartbeat thresholds for the current shift and next critical operating period. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: critical-service register owner, certified or specialist-system owner, CMMS owner and duty manager. Records: current service classification, owner, maintenance/override state, expected heartbeat, dependency list and any certified-status conflict. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: current shift and next critical operating period. Minimum evidence: authoritative critical-service list, expected heartbeat, permission and at least one current owner status channel. Method: calculate freshness, coverage and source agreement; classify each service as confirmed, stale, conflicting or unknown; never impute a safety-critical state. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Never label a service healthy from silence or nearby environmental data; unknown critical status must be deferred to its owner. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: identify which status interfaces or owner confirmations must be checked before prioritising work.

FM-004 RECURRENT FAULT AFTER REPAIR

L3 diagnosis | R2 verified integration

FACILITIES MANAGERS QUESTION

Has this repaired asset returned to the same abnormal pattern under comparable demand?

SENSORS AND TELEMETRY

Proposed. Scope: the repaired asset, its control points, energy or service-output points and verified served spaces. Evidence: pre- and post-repair run state, command/feedback, fault events, load, temperature, pressure, flow, current or vibration features as applicable. Timing: source-native events and 1-minute summaries spanning the pre-fault baseline, repair, handback and current recurrence window. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: CMMS owner, asset owner, competent engineer or contractor handback owner, BMS owner and operating-schedule owner. Records: repair scope, exact intervention and handback times, acceptance criteria, current schedule/load, prior fault signature and unresolved defects. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: diagnosis. Interval: matched pre-repair and post-handback operating windows. Minimum evidence: signed handback, same asset identity, comparable load/regime and valid points across both windows. Method: compare the current signature with the accepted post-repair baseline and earlier fault pattern; separate recurrence, new fault and sensor anomaly hypotheses. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not declare repair failure from unmatched loads or a single point; competent inspection remains necessary. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: decide whether to reopen the prior investigation, seek a different diagnosis or continue monitoring.

1. Asset status, service requests and operational overview

Build a current, evidence-qualified operational picture and decide what needs review, coordination or escalation.

FM-005 OUT-OF-HOURS OPERATION

L2 comparison | R2 verified integration

FACILITIES MANAGERS QUESTION

Which assets are running outside approved schedules without a current recorded exception?

SENSORS AND TELEMETRY

Proposed. Scope: scheduled HVAC, lighting, water, electrical and specialist assets plus their verified served zones. Evidence: run state, power/current, flow, valve/damper state, setpoint or lighting state as appropriate. Timing: source-native state changes and 5-minute summaries across the last seven out-of-hours periods. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

ANALYSIS REQUIRED

Operation: comparison. Interval: current out-of-hours window and a matched recent schedule window. Minimum evidence: verified asset-to-schedule mapping, current owner exception status and valid run-state evidence. Method: compare observed operation with the effective schedule; separate authorised exception, stale schedule, control fault and unknown status. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

AUTHORITATIVE SUPPORTING DATA

Owners: asset and schedule owners, BMS owner, room-use/booking owner and CMMS owner. Records: approved operating schedules, temporary overrides, authorised events, maintenance windows, shutdowns and current owner exceptions. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not infer unauthorised presence or switch equipment off; a schedule conflict is an owner-data issue until confirmed. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare an owner-specific list for schedule correction, control review or documented exception.

FM-006 COORDINATED SITE VISIT

L3 recommendation | R2 verified integration

FACILITIES MANAGERS QUESTION

Which open facilities tasks can be combined into one authorised site visit without mixing incompatible work, access or permits?

SENSORS AND TELEMETRY

Owner-data led; optional proposed telemetry. Scope: tasks, assets and work areas that share a verified location, service path or competent trade. Evidence: optional current asset status and non-identifying area-condition context; no sensing is required for the core grouping. Timing: current status at retrieval and any owner-defined pre-visit check within the proposed work window. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

ANALYSIS REQUIRED

Operation: recommendation. Interval: the planning horizon for the next authorised site visit. Minimum evidence: current open tasks with verified location, trade, dependencies, access and permit status. Method: cluster compatible tasks by place, service and competence; exclude conflicting isolations, permissions, hazards or owner windows; show unresolved prerequisites. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

AUTHORITATIVE SUPPORTING DATA

Owners: CMMS owner, permit-to-work owner, access-control owner, room/service owner, stores/parts owner and contractor coordinator. Records: task scope, trade/competence, exact location, access, permit, isolation, parts, dependencies and proposed visit windows. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not schedule the visit, issue permits or grant access; incompatible or unmapped tasks remain separate. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare a consolidated visit proposal and a clear prerequisite list for the responsible owners.

1. Asset status, service requests and operational overview

Build a current, evidence-qualified operational picture and decide what needs review, coordination or escalation.

FM-007 LOCAL OR SHARED SERVICE ISSUE

L3 diagnosis | R2 verified integration

FACILITIES MANAGERS QUESTION

Is this problem confined to one room, one served zone or a wider building service?

SENSORS AND TELEMETRY

Proposed. Scope: the reported room, comparable rooms on the same verified served zone, terminal unit and upstream service path. Evidence: room environmental values, terminal command/feedback, branch flow/pressure/current and upstream plant state relevant to the symptom. Timing: one-minute aligned trends across the complaint window and one matched normal window. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: room/service-topology owner, CMMS owner, BMS or specialist-system owner and room-use owner. Records: current room purpose, served-zone mapping, request history, operating schedule, overrides and recent work. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: diagnosis. Interval: complaint window plus matched room, zone and building comparators. Minimum evidence: valid room-level evidence, verified service graph and at least one eligible comparator at each tested level. Method: test local, zone and upstream patterns in that order; retain direct observations separately from inferred scope. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not substitute corridor or adjacent-room data for the reported room or claim root cause without corroboration. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: choose the narrowest defensible inspection scope and avoid unnecessary building-wide escalation.

FM-008 UNPRIORITISABLE REQUESTS

L2 calculation | R1 owner-data ready

FACILITIES MANAGERS QUESTION

Which open requests cannot yet be prioritised responsibly, and what minimum evidence is missing?

SENSORS AND TELEMETRY

Owner-data led; optional proposed telemetry. Scope: all open facilities requests and the assets, rooms or services each request claims to affect. Evidence: optional point-health and mapping status only; the core answer is an authoritative-data completeness audit. Timing: current request state at retrieval and owner-defined age thresholds. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: CMMS/service-request owner, asset/room register owners and relevant service owners. Records: request location, affected service, consequence, owner, duplicate links, access constraints, current status and evidence attachments. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: current prioritisation cycle. Minimum evidence: open request plus minimum mandatory fields and a verified asset/room/service reference. Method: apply a published evidence checklist; classify ready, clarification needed, owner-deferred or not assessable; identify the smallest missing item. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not invent severity, infer people affected or downgrade an authorised exception; missing fields stay visible. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: send precise clarification requests and prevent weak data from driving the work queue.

1. Asset status, service requests and operational overview

Build a current, evidence-qualified operational picture and decide what needs review, coordination or escalation.

FM-009 MONITORING HEALTH PRIORITIES

L3 recommendation | R2 verified integration

FACILITIES MANAGERS QUESTION

Which monitoring points should be checked first because current facilities decisions depend on them?

SENSORS AND TELEMETRY

Proposed. Scope: points supporting open decisions, their mapped assets/rooms and any independent reference or redundant point. Evidence: heartbeat, communications quality, range/flat-line/drift indicators, calibration or functional-check status and recent maintenance state. Timing: owner-defined heartbeat plus 1-minute summaries across the decision window and previous healthy baseline. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: sensor/IoT owner, BMS or specialist-system owner, asset owner and CMMS owner. Records: point inventory, mapping, expected cadence, commissioning/calibration, maintenance, known defects and decision dependencies. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: recommendation. Interval: current shift and open-decision horizon. Minimum evidence: current decision list, verified point dependencies, expected cadence and health criteria. Method: risk-rank points by decision consequence, evidence centrality, staleness, drift and lack of redundancy; distinguish repair, validation and mapping needs. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not declare the underlying asset faulty because its monitor is unhealthy; critical decisions require owner confirmation. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: target validation effort where a bad point would most distort facilities priorities.

FM-010 HANDOVER CHANGE SUMMARY

L2 comparison | R2 verified integration

FACILITIES MANAGERS QUESTION

What material asset, service or evidence changes have occurred since the last signed facilities handover?

SENSORS AND TELEMETRY

Proposed. Scope: assets and services named in either handover plus relevant status, fault and monitoring-health points. Evidence: source-native events and owner-defined summaries from the signed cut-off to current retrieval. Timing: source-native events and current owner summaries from the signed cut-off to current retrieval. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: handover owner, CMMS owner, asset register owner, BMS/specialist-system owners and incident/closure owners. Records: signed handovers, work-order state, asset/point mapping changes, overrides, closures, incidents and monitoring-health changes. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: last signed handover to current retrieval. Minimum evidence: two signed/versioned handover states and current owner records for changed items. Method: perform a versioned change set; group new, resolved, worsened, transferred and evidence-degraded items; preserve conflicting owner states. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not treat an unsigned note as authoritative or convert change detection into an instruction to operate or authorise. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare a concise shift briefing and identify the smallest owner checks still needed.

2. Space use, occupancy patterns and service demand

Use anonymous aggregate demand evidence to prepare services without identifying people or treating occupancy as permission.

FM-011 PEAK SERVICE DEMAND

L3 forecast | R2 verified integration

FACILITIES MANAGERS QUESTION

For the next teaching or event peak, which areas are most likely to need extra facilities checks or support?

SENSORS AND TELEMETRY

Proposed. Scope: authorised event and teaching areas, adjacent public routes and linked amenities within commissioned anonymous-count coverage. Evidence: anonymous doorway counts, zone occupancy totals, CO2 as ventilation context, lift/door status and amenity demand indicators where approved. Timing: 1-5 minute aggregate counts and environmental summaries over comparable peaks; no identities or movement histories. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

ANALYSIS REQUIRED

Operation: forecast. Interval: the next declared peak from pre-arrival through dispersal. Minimum evidence: current schedule and owner status plus at least three comparable events or an explicitly low-confidence model. Method: forecast aggregate demand by area and service; adjust for room type, event scale, day/time and known closures; publish prediction intervals. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

AUTHORITATIVE SUPPORTING DATA

Owners: timetable/event owners, room and opening-hours owners, service/amenity owners, accessibility owner and facilities duty owner. Records: current schedule, expected scale, room capacity/purpose, service rounds, closures and approved contingency arrangements. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not infer attendance, identify people or treat predicted demand as permission, staffing adequacy or safety certification. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: plan inspections, replenishment and owner coordination before the peak.

FM-012 CROWDING AND UNDERUSE

L2 comparison | R2 verified integration

FACILITIES MANAGERS QUESTION

Are some shared areas repeatedly congested while eligible nearby alternatives remain lightly used?

SENSORS AND TELEMETRY

Proposed. Scope: shared areas and only those alternatives with current matching purpose, opening, access and service status. Evidence: anonymous area counts, doorway totals and numerical acoustic/environmental context where commissioned. Timing: 1-5 minute aggregates during matched schedule windows over at least four comparable days. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

ANALYSIS REQUIRED

Operation: comparison. Interval: matched peak windows across eligible spaces. Minimum evidence: verified comparable spaces, adequate aggregate coverage and current owner status. Method: compare occupancy ratio, duration above owner-defined crowding bands and service conditions; adjust for room type and scheduled demand. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

AUTHORITATIVE SUPPORTING DATA

Owners: room/space owner, booking/timetable owner, access and opening-hours owners, accessibility owner and facilities service owners. Records: current purpose, capacity, eligible use, access, closures, schedules and service availability. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not redirect users, infer who is present or treat low counts as availability, access or suitability. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: identify where facilities coordination or a space-owner discussion may reduce recurring pressure.

2. Space use, occupancy patterns and service demand

Use anonymous aggregate demand evidence to prepare services without identifying people or treating occupancy as permission.

FM-013 AMENITY DEMAND ALIGNMENT

L3 forecast | R2 verified integration

FACILITIES MANAGERS QUESTION

Do predicted amenity peaks align with the next authorised replenishment and service rounds?

SENSORS AND TELEMETRY

Proposed. Scope: approved public amenities and their defined service catchments. Evidence: anonymous footfall or doorway totals, dispenser/fill/flow indicators and service-point status where commissioned. Timing: event-driven device status and 5-15 minute aggregates across comparable service periods. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: amenity service owner, cleaning/waste owner, event/timetable owner, stores owner and facilities duty owner. Records: service schedule, stock thresholds, approved opening, event demand, closures and last completed round. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: forecast. Interval: next service round and following demand peak. Minimum evidence: current service status, valid aggregate demand evidence and at least one comparable operating period. Method: forecast service-point demand and time-to-threshold; show owner data separately from sensor estimates and include uncertainty from missing coverage. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not infer individual use, cleanliness or staffing need; an estimate cannot replace a service inspection. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: bring forward, retain or defer a replenishment check with a stated confidence level.

FM-014 ENTRANCE AND ROUTE SUPPORT

L3 forecast | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Which entrances or route segments need a facilities inspection before the next high-footfall transition?

SENSORS AND TELEMETRY

Proposed. Scope: current public/authorised entrances, route segments, doors, lifts and any temporary diversion. Evidence: anonymous directional counts, door/lift status, numerical surface/weather indicators and owner-issued closure feeds. Timing: 1-minute aggregate counts around comparable transitions plus current asset status and official weather window. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: event/timetable owner, route/wayfinding owner, access/security owner, lift/door owners, accessibility owner and facilities duty owner. Records: current opening, authorised routes, temporary works, closures, lift/door status, event timing and inspection criteria. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: forecast. Interval: pre-transition inspection window through peak dispersal. Minimum evidence: current route authority plus valid coverage on candidate segments and comparable transitions. Method: forecast route load and combine it with owner-defined asset/weather inspection triggers; show uncovered segments as unknown. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not disclose security-sensitive details, grant access or certify route accessibility/safety; owner routes and closures always prevail. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: target a pre-peak inspection of the entrances and route assets most likely to affect service.

2. Space use, occupancy patterns and service demand

Use anonymous aggregate demand evidence to prepare services without identifying people or treating occupancy as permission.

FM-015 CONDITIONS UNDER HIGH USE

L2 comparison | R2 verified integration

FACILITIES MANAGERS QUESTION

Under which verified high-use conditions do room-environment complaints recur most consistently?

SENSORS AND TELEMETRY

Proposed. Scope: complaint rooms with valid room-level sensors, verified served zones and current room-use context. Evidence: room temperature, RH, CO2, PM where appropriate, anonymous occupancy, supply airflow and relevant control state. Timing: 1-minute summaries over complaint windows and matched high-use/non-complaint sessions. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: room-use owner, booking/timetable owner, BMS owner, environmental-sensor owner, weather source and CMMS owner. Records: complaint time, current room purpose, schedule, occupancy context, served-zone mapping, overrides and maintenance. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: complaint sessions versus matched comparable sessions. Minimum evidence: valid room-level coverage, aligned clocks and at least two eligible comparator periods. Method: compare complaint and non-complaint conditions after matching room, load, duration, weather and control regime; report association only. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not infer individual comfort, health or attendance, and do not substitute corridor data for the room. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: identify repeatable conditions that justify a focused facilities or competent-engineer review.

FM-016 OUT-OF-HOURS SPACE DEMAND

L2 comparison | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Which areas show sustained aggregate use beyond normal facilities-service windows?

SENSORS AND TELEMETRY

Proposed. Scope: approved aggregate zones with current opening/access definitions; no person-level traces. Evidence: anonymous area counts, doorway totals and lighting/HVAC state where commissioned. Timing: 5-minute aggregates with no persistent device identifiers across at least four weeks. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: opening-hours owner, space owner, access/security owner, timetable/event owner and facilities-service owners. Records: current opening/access rules, authorised events, service windows, closures and exceptional-use records. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: defined out-of-hours periods versus matched in-hours periods. Minimum evidence: current policy window, valid aggregate coverage and authorised exceptional-use context. Method: calculate frequency and duration of aggregate use by zone; separate observed use from inferred service demand and publish coverage limits. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not identify or track occupants, infer unauthorised activity or change access/service policy. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: review whether service windows or owner coordination need evidence-led discussion.

2. Space use, occupancy patterns and service demand

Use anonymous aggregate demand evidence to prepare services without identifying people or treating occupancy as permission.

FM-017 CURRENT USE VERSUS OFFICIAL PURPOSE

L2 comparison | R2 verified integration

FACILITIES MANAGERS QUESTION

Does aggregate use of this space still fit its current recorded purpose and facilities-service assumptions?

SENSORS AND TELEMETRY

Proposed. Scope: the named space and only current eligible comparator spaces. Evidence: anonymous occupancy duration, doorway totals, environmental load and equipment-energy context where commissioned. Timing: 5-minute aggregates across representative weeks, excluding outages and mapping-change periods. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: space/room owner, room register owner, booking/timetable owner, accessibility owner and service owners. Records: current official purpose, capacity, eligible activities, service assumptions, opening and planned changes. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: current term or reporting period. Minimum evidence: current official purpose plus adequate aggregate coverage across representative operating periods. Method: compare observed aggregate patterns with the recorded use envelope; flag sustained mismatch without classifying activities or people. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not infer the activity taking place, declare misuse or change room purpose. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare evidence for a room-owner and service-planning review.

FM-018 HIGH-CAPACITY LOW-USE ROOMS

L2 calculation | R2 verified integration

FACILITIES MANAGERS QUESTION

Which officially allocated high-capacity rooms show persistently low aggregate use under comparable schedules?

SENSORS AND TELEMETRY

Proposed. Scope: eligible rooms with current purpose, capacity, access and commissioned aggregate counting. Evidence: anonymous doorway counts or room occupancy totals; environmental evidence only as service context. Timing: 5-minute aggregates across matched scheduled sessions and at least one full representative term segment. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: room register owner, timetable/booking owners, space owner, access owner and facilities-service owners. Records: current capacity, purpose, scheduled activity, cancellations, closures, equipment status and counting coverage. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: representative scheduled periods with comparable activity type. Minimum evidence: current capacity/schedule plus adequate aggregate coverage and cancellation data. Method: calculate utilisation ranges and confidence intervals; exclude cancelled, inaccessible, closed or sensor-fault periods; avoid individual attendance claims. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not identify participants, judge teaching demand or recommend room reallocation without room/timetabling owners. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: identify rooms whose facilities-service assumptions merit review, not automatic reassignment.

2. Space use, occupancy patterns and service demand

Use anonymous aggregate demand evidence to prepare services without identifying people or treating occupancy as permission.

FM-019 TEMPORARY REST OR SEATING SUPPORT

L3 recommendation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Where could temporary rest or waiting support reduce congestion without obstructing an authorised route?

SENSORS AND TELEMETRY

Proposed. Scope: currently authorised public areas, route clearances, doors, lift landings and candidate furniture zones. Evidence: anonymous counts and numerical acoustic/environmental context; no sensing can establish permission or accessibility. Timing: 1-minute aggregate counts during comparable peaks plus current owner status. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: space owner, route/accessibility owner, fire/safety authority, event owner, facilities duty owner and furniture/service owner. Records: current route widths/clearances, approved use, furniture rules, event schedule, closures and accessibility requirements. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: recommendation. Interval: the declared event or service period. Minimum evidence: authorised candidate locations plus valid demand evidence and competent-owner clearance requirements. Method: rank only eligible candidate areas by expected congestion relief, route impact and service practicality; show uncertainty and no-sensor alternatives. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not place furniture, certify accessibility/egress or treat quietness/low use as availability or permission. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare a short list for competent-owner review before any temporary setup.

FM-020 UPCOMING EVENT SERVICE COVERAGE

L3 recommendation | R2 verified integration

FACILITIES MANAGERS QUESTION

Which facilities services need owner verification before this high-demand event?

SENSORS AND TELEMETRY

Proposed. Scope: event venue, verified service paths, public routes and linked amenities. Evidence: room environmental status, lift/door state, utility/plant status, anonymous aggregate demand and amenity indicators where commissioned. Timing: current status plus comparable-event aggregates and owner-defined pre-event checks. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: event organiser, room owner, facilities service owners, BMS/utility owners, accessibility owner, security and cleaning/waste owners. Records: event schedule/scale, official venue, service requirements, setup/handback, closures, contingency and owner acceptance criteria. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: recommendation. Interval: pre-event readiness window through handback. Minimum evidence: current event brief, named service owners and valid status for each required service. Method: build an owner-by-owner readiness matrix; distinguish authoritative status, telemetry context and unresolved evidence; prioritise checks by consequence. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not declare the event ready, grant access or replace organiser, security, safety or service-owner sign-off. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: coordinate the smallest set of owner checks needed before the event.

3. Temperature, air, lighting, acoustics and comfort

Qualify room and served-zone environmental evidence before inspection, diagnosis, seasonal preparation or intervention review.

FM-021 CURRENT ENVIRONMENTAL EXCEPTIONS

L2 calculation | R2 verified integration

FACILITIES MANAGERS QUESTION

Which occupied rooms show persistent temperature, CO2 or particulate exceptions supported by valid room-level evidence?

SENSORS AND TELEMETRY

Proposed. Scope: rooms with current room-level sensor placement and verified served-zone mapping. Evidence: temperature, RH, direct NDIR CO2, PM2.5/PM10 where appropriate, anonymous occupancy and supply-air context. Timing: 1-minute records with owner-defined persistence windows; exclude commissioning, maintenance and known sensor-fault periods. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: environmental-sensor owner, BMS owner, room-use owner, weather source, CMMS owner and policy/criterion owner. Records: current room purpose, schedule, served zone, sensor placement/calibration, operating state, complaints and applicable operational criteria. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: current occupied period and recent comparable sessions. Minimum evidence: valid room-level coverage, current room use, healthy sensors and an explicit operational criterion. Method: calculate duration and magnitude of exceptions; retain measured variables separately and avoid composite health claims. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not diagnose health, comfort or compliance; corridor data is not room evidence. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prioritise rooms for facilities inspection or competent ventilation/thermal review.

FM-022 COMFORT COMPLAINT EVIDENCE

L2 comparison | R2 verified integration

FACILITIES MANAGERS QUESTION

What does the evidence support about this room-comfort report, and what remains unknown?

SENSORS AND TELEMETRY

Proposed. Scope: the reported room, valid room-level comfort/environment points, served zone and eligible comparator rooms. Evidence: temperature, RH, globe/radiant temperature and air speed where installed, CO2, anonymous occupancy, airflow and outdoor weather. Timing: 1-minute records over the report window plus matched comparator periods. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: room-use owner, sensor/BMS owners, weather source, CMMS/complaint owner and any approved comfort criterion owner. Records: report time/location, current room use, sensor placement/health, occupancy context, control state and recent changes. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: reported period versus matched room and weather conditions. Minimum evidence: valid room-level evidence and at least one eligible comparator; advanced comfort variables where a comfort model is claimed. Method: compare measured conditions and variability; separate direct observations from model-based comfort estimates and unknown personal factors. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not infer an individual's comfort, health or productivity or declare a standards outcome without the required variables. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: decide whether the report warrants local inspection, wider zone review or more evidence.

3. Temperature, air, lighting, acoustics and comfort

Qualify room and served-zone environmental evidence before inspection, diagnosis, seasonal preparation or intervention review.

FM-023 RECURRING WARM-UP OR OVERHEATING

L3 diagnosis | R2 verified integration

FACILITIES MANAGERS QUESTION

Which rooms repeatedly warm too slowly at opening or overheat later under matched conditions?

SENSORS AND TELEMETRY

Proposed. Scope: rooms with valid temperature points, served terminal units, upstream plant and relevant facade/weather context. Evidence: room temperature, radiant/surface temperature where available, supply temperature/airflow, valve/damper state, occupancy and solar/weather inputs. Timing: 1-minute records from pre-opening through occupied periods across matched days. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: BMS/plant owner, room-use/timetable owner, sensor owner, weather source and CMMS owner. Records: current schedule, setpoints, start/stop strategy, served-zone mapping, overrides, maintenance and facade orientation where authoritative. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: diagnosis. Interval: matched opening and peak periods across representative weather days. Minimum evidence: healthy room and control points, current schedule and at least three comparable days. Method: compare warm-up slope, peak temperature, load and control response; rank schedule, capacity, solar/load and sensor explanations without proving cause. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not change setpoints or schedules or infer comfort; unmatched weather/load periods stay qualified. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: target commissioning, schedule or local heat-gain checks before the next seasonal peak.

FM-024 HIGH CO2 AND AIRFLOW

L3 diagnosis | R2 verified integration

FACILITIES MANAGERS QUESTION

Are recurring high-CO2 periods associated with low supply airflow or particular aggregate-use conditions?

SENSORS AND TELEMETRY

Proposed. Scope: the named room, its CO2/occupancy points, terminal airflow and verified upstream ventilation path. Evidence: direct NDIR CO2, anonymous occupancy, supply/return airflow, damper/fan state, door/window state where authorised and outdoor CO2 reference. Timing: 30-60 second sampling aggregated to 1 minute across repeated sessions. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: sensor owner, BMS/ventilation owner, room-use owner, timetable/booking owner, weather/outdoor reference owner and CMMS owner. Records: room volume/purpose, current schedule, ventilation design/sequence, sensor placement/calibration, overrides and maintenance. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: diagnosis. Interval: repeated high-CO2 windows plus matched normal sessions. Minimum evidence: valid room CO2 and occupancy evidence, verified airflow mapping and aligned clocks. Method: analyse accumulation and recovery, occupancy-normalised CO2 and control/airflow sequence; report associations and lag uncertainty. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not infer exact attendance, infection risk, health or legal compliance, and do not substitute adjacent-room data. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: decide whether to inspect room sensing, terminal airflow, controls or wider ventilation service.

3. Temperature, air, lighting, acoustics and comfort

Qualify room and served-zone environmental evidence before inspection, diagnosis, seasonal preparation or intervention review.

FM-025 LIGHTING SUITABILITY EXCEPTIONS

L2 comparison | R2 verified integration

FACILITIES MANAGERS QUESTION

Which rooms have lighting levels or control states that do not fit their current authorised task and schedule?

SENSORS AND TELEMETRY

Proposed. Scope: task-relevant measurement zones, lighting circuits/controllers and daylight reference points. Evidence: calibrated illuminance at representative task plane, lighting state/power, occupancy aggregate and outdoor/daylight context. Timing: 10-30 second sampling aggregated to 1 minute over scheduled use and matched daylight periods. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: lighting-control owner, room-use/booking owner, room/equipment owner, accessibility owner and sensor owner. Records: current task/purpose, schedule, lighting design/criterion, control sequence, overrides, maintenance and daylight orientation. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: current or reported task window versus matched room/daylight periods. Minimum evidence: representative task-plane coverage, current room task and healthy lighting/control points. Method: compare illuminance distribution and control state against the owner-defined task requirement; separate sensor-placement, glare and control explanations. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not certify visual accessibility or compliance, infer activity or alter lighting controls. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prioritise lamp/control inspection or a competent lighting assessment.

FM-026 BUILDING-SERVICES NOISE

L3 diagnosis | R2 verified integration

FACILITIES MANAGERS QUESTION

Do repeated numerical noise reports align with operation of nearby plant, fans, pumps, lifts or other services?

SENSORS AND TELEMETRY

Proposed. Scope: reported room/zone, candidate assets and verified acoustic/service paths. Evidence: privacy-preserving numerical LAeq/percentile levels only, equipment run state, speed/current and vibration features; no audio recording. Timing: 1-second local sound calculation stored as 1-minute statistics plus source-native equipment events. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: acoustic-sensor owner, asset/BMS owners, room-use owner, lift/specialist owner and CMMS/report owner. Records: report time/location, asset-service mapping, equipment schedules, maintenance, room use and approved acoustic criteria where applicable. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: diagnosis. Interval: report windows plus matched asset-on/off and normal periods. Minimum evidence: valid numerical acoustic coverage, aligned asset events and verified candidate mapping. Method: compare time-aligned sound statistics with equipment states; rank candidate services and non-service background without identifying speech. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Never record or analyse speech, infer conversations or certify nuisance/compliance; association does not prove source. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: choose which asset or acoustic path merits competent inspection.

3. Temperature, air, lighting, acoustics and comfort

Qualify room and served-zone environmental evidence before inspection, diagnosis, seasonal preparation or intervention review.

FM-027 HUMIDITY OR PARTICULATE SOURCE SCOPE

L3 diagnosis | R2 verified integration

FACILITIES MANAGERS QUESTION

Is this humidity or particulate anomaly local, outdoor-driven or shared across a verified ventilation zone?

SENSORS AND TELEMETRY

Proposed. Scope: affected room, same-zone comparators, supply/return path and outdoor reference. Evidence: RH, temperature, PM2.5/PM10 where appropriate, supply/return conditions, filtration pressure and outdoor PM/weather. Timing: 30-60 second sampling aggregated to 1 minute across onset, peak and recovery. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: sensor owner, ventilation/BMS owner, outdoor reference owner, room-use owner and CMMS owner. Records: sensor placement/calibration, zone mapping, filtration/ventilation state, cleaning/work events, weather and current room use. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: diagnosis. Interval: anomaly window plus matched normal and outdoor conditions. Minimum evidence: healthy room and outdoor sensors, verified zone mapping and aligned ventilation state. Method: compare spatial gradients, indoor/outdoor ratios and event sequence; rank local source, outdoor ingress, shared ventilation and sensor explanations. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not identify a chemical source, make health claims or use uncalibrated low-cost values as compound-specific evidence. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: direct the first inspection to the room, ventilation zone, filtration path or sensor itself.

FM-028 ENVIRONMENTAL DATA FITNESS

L3 recommendation | R2 verified integration

FACILITIES MANAGERS QUESTION

Which environmental sensors are unfit for room-level facilities decisions, and what would restore confidence?

SENSORS AND TELEMETRY

Proposed. Scope: all environmental points used in current facilities questions, with exact room, mounting and served-zone mapping. Evidence: heartbeat, missingness, range, flat-line, drift, cross-sensor agreement, calibration/functional-check state and relocation history. Timing: quality metrics over each decision window plus 30-day health context. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: sensor/IoT owner, BMS owner, calibration/commissioning owner, asset/room register owner and CMMS owner. Records: point inventory, model/specification, placement, calibration, maintenance, firmware/configuration and decision dependencies. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: recommendation. Interval: current decision set and next validation window. Minimum evidence: current mapping, expected cadence, quality criteria and at least one independent reference where consequence is high. Method: score fitness by mapping, placement, freshness, completeness, plausibility and reference agreement; prescribe validate, recalibrate, relocate, repair or retire. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not use adjacent-room data as a silent substitute or diagnose the room from an unfit point. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: create a prioritised sensor-validation plan linked to facilities decision risk.

3. Temperature, air, lighting, acoustics and comfort

Qualify room and served-zone environmental evidence before inspection, diagnosis, seasonal preparation or intervention review.

FM-029 CONTROL-CHANGE OUTCOME

L2 comparison | R2 verified integration

FACILITIES MANAGERS QUESTION

Did the recent control change improve the target rooms without creating a measurable problem elsewhere?

SENSORS AND TELEMETRY

Proposed. Scope: target rooms, affected control points, upstream assets and pre-defined spillover zones. Evidence: room environmental points, control commands/feedback, energy/service-output points and aggregate occupancy/weather context. Timing: 1-minute records across matched pre-change and post-stabilisation periods. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: change-control owner, BMS owner, room-use owner, sensor owner, weather source and CMMS owner. Records: approved change scope/time, acceptance criteria, affected assets/zones, overrides, maintenance and known confounders. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: matched pre-change and post-change operating windows. Minimum evidence: signed change record, stable mapping and sufficient valid data before and after under comparable conditions. Method: compare service outcome, variability, energy and spillover indicators; normalise for load/weather and show sensitivity to exclusions. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not claim causality, approve the change or operate controls; periods with maintenance or override remain separate. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: decide whether to retain, investigate or seek competent review of the control change.

FM-030 SEASONAL CONDITIONING PRIORITIES

L3 recommendation | R2 verified integration

FACILITIES MANAGERS QUESTION

Which served zones should be verified first before the next heating or cooling changeover?

SENSORS AND TELEMETRY

Proposed. Scope: served zones, terminal units, upstream plant and representative perimeter/internal rooms. Evidence: room temperature/RH, supply/return temperature, airflow/pressure, valve/damper state, plant run state and weather context. Timing: 1-minute operational summaries over recent shoulder-season and prior changeover periods. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: seasonal plan owner, BMS/plant owner, room-use owner, sensor owner, weather source and CMMS owner. Records: changeover date, commissioning checks, zone criticality, known complaints/faults, overrides and prior seasonal performance. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: recommendation. Interval: pre-changeover planning window. Minimum evidence: current service topology, healthy points and representative prior seasonal evidence. Method: risk-rank zones by unresolved defects, poor prior response, sensor gaps, high consequence and limited contingency; state confidence. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not start plant, change controls or certify readiness; competent owners approve seasonal operation. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: plan a focused verification sequence before seasonal conditions arrive.

4. Cleaning, waste, amenities and public-facing services

Coordinate inspections and service rounds using owner status first and proportionate non-identifying telemetry as context.

FM-031 USE-LED CLEANING CHECK

L3 recommendation | R2 verified integration

FACILITIES MANAGERS QUESTION

Which shared or public areas merit an earlier cleaning inspection today?

SENSORS AND TELEMETRY

Proposed. Scope: approved shared/public areas and their service catchments. Evidence: anonymous aggregate counts, doorway totals, bin/fill indicators and numerical environmental context where proportionate. Timing: 5-15 minute aggregates since the last completed service round. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: cleaning-service owner, room/space owner, event/timetable owner and facilities duty owner. Records: last completed round, current service standard, event/use schedule, closures, incident reports and owner exceptions. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: recommendation. Interval: last completed round to next planned service. Minimum evidence: current service status plus valid aggregate use evidence for the named area. Method: rank inspection need by aggregate turnover, event intensity, time since service and owner-reported issue; keep sensor estimate separate from cleanliness status. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not infer hygiene, individual behaviour or automatically reschedule cleaning; inspection remains the service owner's decision. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: identify which areas should be checked before the next routine round.

FM-032 CONSUMABLE REPLENISHMENT RISK

L2 estimate | R2 verified integration

FACILITIES MANAGERS QUESTION

Which public amenities may need a replenishment check before the next service round?

SENSORS AND TELEMETRY

Proposed. Scope: approved dispensers or service points with verified device-to-location mapping. Evidence: owner-issued stock status, dispenser level/weight or flow indicators and anonymous aggregate demand where commissioned. Timing: event-driven level changes or 5-15 minute summaries since the last refill. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: amenity-service owner, stores owner, cleaning/service-round owner and facilities duty owner. Records: last refill, product threshold, approved opening, expected demand, device maintenance and current closure status. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: estimate. Interval: current time to next planned round. Minimum evidence: current owner status or calibrated level indicator plus valid demand history. Method: estimate time-to-threshold with uncertainty; distinguish measured level, model estimate and unknown device state. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not infer cleanliness, individual use or staffing need, and do not claim a refill occurred without owner confirmation. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: bring forward a physical check or retain the planned round with an evidence-based reason.

4. Cleaning, waste, amenities and public-facing services

Coordinate inspections and service rounds using owner status first and proportionate non-identifying telemetry as context.

FM-033 WASTE COLLECTION PRIORITY

L3 recommendation | R2 verified integration

FACILITIES MANAGERS QUESTION

Which approved waste points should be inspected first on the next collection round?

SENSORS AND TELEMETRY

Proposed. Scope: approved waste points, their access routes and service catchments. Evidence: fill level/weight, temperature only where justified, anonymous nearby aggregate use and device-health status. Timing: 5-15 minute summaries and event-driven threshold alerts across the current collection cycle. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: waste-service owner, space owner, event/timetable owner, access owner and facilities duty owner. Records: collection schedule, waste-stream type, approved capacity/threshold, closures, contamination reports and last service. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: recommendation. Interval: current collection cycle. Minimum evidence: current service status, verified device mapping and either valid fill evidence or owner report. Method: rank physical inspection by predicted fill, service consequence, stream restrictions and route practicality; show unknown or conflicting states. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not classify waste contents, identify users or authorise hazardous-waste handling; specialist streams stay owner-controlled. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: sequence the collection inspection route and escalate restricted streams correctly.

FM-034 ODOUR INVESTIGATION SCOPE

L3 diagnosis | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

What evidence should guide the first facilities inspection after a reported odour?

SENSORS AND TELEMETRY

Proposed. Scope: reported location, nearby drains/amenities, verified ventilation zone and candidate service assets. Evidence: indicative TVOC only, CO2, RH, temperature, drain/water status, airflow/extract state and outdoor reference where applicable. Timing: 30-60 second sampling aggregated to 1 minute around the report and recovery. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: report/CMMS owner, ventilation/BMS owner, water/drainage owner, cleaning owner, room-use owner and safety owner. Records: report time/location, room purpose, recent cleaning/work/activity records, service states, sensor limitations and hazard procedures. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: diagnosis. Interval: report window and immediate inspection period. Minimum evidence: authoritative report, valid mapped points and current safety/incident status. Method: compare spatial pattern and service-event timing; rank ventilation, drainage, cleaning, outdoor and sensor explanations without identifying chemicals. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not identify a substance or declare the area safe from generic TVOC; suspected hazardous odours require competent safety procedures. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: choose a safe first inspection path and the correct owner escalation.

4. Cleaning, waste, amenities and public-facing services

Coordinate inspections and service rounds using owner status first and proportionate non-identifying telemetry as context.

FM-035 WEATHER-RELATED ENTRANCE CONDITION

L3 recommendation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Which public entrances need inspection after the current rain, wind or cold-weather event?

SENSORS AND TELEMETRY

Proposed. Scope: current authorised entrances, external approaches, matting/drainage zones, doors and adjacent public floor zones. Evidence: official weather, surface moisture or temperature where commissioned, door state and anonymous aggregate use. Timing: event-driven weather/door state plus 1-5 minute surface summaries during and after the event. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: entrance/space owner, cleaning owner, security/access owner, weather source, accessibility owner and facilities duty owner. Records: current public entrances, inspection criteria, cleaning response, closures, temporary routes and event/opening schedule. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: recommendation. Interval: current weather event through post-event inspection window. Minimum evidence: authorised entrance list, current weather and either valid surface evidence or owner report. Method: rank entrances by exposure, service demand, known drainage/matting issues and evidence confidence; keep uncovered approaches unknown. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not certify slip safety, open/close entrances or override security/accessibility routes. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: target timely physical inspections and coordinate cleaning or owner response.

FM-036 ABNORMAL WATER USE

L3 diagnosis | R2 verified integration

FACILITIES MANAGERS QUESTION

Does the current water pattern justify a leak or fixture inspection, or fit an authorised high-demand activity?

SENSORS AND TELEMETRY

Proposed. Scope: whole-building and branch meters, candidate fixtures/plant and verified service catchments. Evidence: flow, pressure, meter pulses, leak sensors and relevant plant state; anonymous counts only where proportionate. Timing: 1-minute flow totals with event-driven leak status across current and matched operating periods. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: water-meter owner, plumbing/plant owner, room/activity owner, event/cleaning owner and CMMS owner. Records: meter hierarchy, normal baselines, authorised high-use activity, maintenance, closures and known meter changes. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: diagnosis. Interval: current anomaly window plus matched day/time and weather/activity periods. Minimum evidence: stable meter boundary, aligned clocks, current activity context and adequate downstream visibility. Method: apply change-point and night-flow checks; compare branch patterns; rank leak, legitimate demand, meter fault and unknown distribution. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not declare a leak from whole-building demand alone, identify users or isolate water. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: decide where a competent plumbing inspection should start and whether evidence is insufficient.

4. Cleaning, waste, amenities and public-facing services

Coordinate inspections and service rounds using owner status first and proportionate non-identifying telemetry as context.

FM-037 DRINKING-WATER POINT STATUS

L2 calculation | R1 owner-data ready

FACILITIES MANAGERS QUESTION

Which drinking-water points are officially available now, and what verified alternative can I communicate?

SENSORS AND TELEMETRY

Owner-data led; optional proposed telemetry. Scope: owner-approved drinking-water points and any non-safety operational status feed. Evidence: optional flow/pressure and device heartbeat for service context; no generic sensor can certify water quality. Timing: current owner status at retrieval and owner-defined heartbeat if available. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: drinking-water/service owner, water-quality competent owner, room/access owner and facilities communications owner. Records: current availability, maintenance/closure, sampling or quality assurance status where authorised, accessibility and approved alternatives. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: current public-service period. Minimum evidence: owner-issued availability for each point and a current approved alternative list. Method: merge owner statuses without overriding conflicts; label telemetry as context and stale owner records as unresolved. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not certify potability, accessibility or safety from flow/pressure, and do not publish restricted quality records. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: communicate a verified available point or clearly defer when status is unknown.

FM-038 PUBLIC AMENITY CONDITION PATTERN

L2 comparison | R2 verified integration

FACILITIES MANAGERS QUESTION

Which public seating or amenity areas repeatedly miss their approved environmental service targets?

SENSORS AND TELEMETRY

Proposed. Scope: approved public seating/amenity areas with valid room/zone mapping. Evidence: temperature, RH, CO2, numerical acoustic level and lighting where relevant, plus anonymous aggregate use. Timing: 1-minute summaries across representative service periods. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: space/amenity owner, sensor/BMS owners, accessibility owner, event/timetable owner and CMMS owner. Records: current purpose, service target, opening, schedule, complaints, maintenance and sensor placement/health. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: representative periods across comparable days/events. Minimum evidence: current owner target, valid area-level coverage and comparable operating conditions. Method: compare duration and variability against owner-defined targets after matching demand and weather; report only measured dimensions. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not certify hygiene, accessibility, comfort or readiness, and do not infer individual experience. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: identify amenities needing inspection, service adjustment review or better evidence.

4. Cleaning, waste, amenities and public-facing services

Coordinate inspections and service rounds using owner status first and proportionate non-identifying telemetry as context.

FM-039 TEMPORARY SIGNAGE OR BARRIER NEED

L3 recommendation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Where does an authorised facilities issue require a temporary signage or barrier review?

SENSORS AND TELEMETRY

Proposed. Scope: owner-declared issue boundary, affected route segments, doors, assets and authorised diversion. Evidence: owner-issued fault/closure status, door/lift state and anonymous route demand only as context. Timing: current status and short forecast across the declared issue window. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: issue owner, facilities duty owner, route/accessibility owner, security, fire/safety authority and communications owner. Records: authorised issue scope, current routes/closures, barrier/signage standard, event schedule and expiry/inspection rules. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: recommendation. Interval: declared issue window. Minimum evidence: owner-confirmed issue, route boundary and competent clearance requirements. Method: identify candidate sign/barrier locations for owner review based on route clarity, demand and accessibility constraints; keep uncovered paths unknown. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not place barriers, close routes, certify egress/accessibility or override security/safety instructions. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare a competent-owner review of temporary communication and route protection.

FM-040 EVENT CLEANING HANDOVER

L2 calculation | R2 verified integration

FACILITIES MANAGERS QUESTION

Is the evidence complete enough for organiser and service owners to review event-space handover?

SENSORS AND TELEMETRY

Owner-data led; optional proposed telemetry. Scope: event rooms, linked public areas, approved waste points and amenities. Evidence: optional aggregate use, fill indicators and room environmental status; owner completion records are primary. Timing: event period through declared handover time. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: event organiser, room owner, cleaning/waste owners, facilities duty owner and incident/maintenance owners. Records: event schedule, setup/handback criteria, service completion, damage/issue reports, closures and owner acceptance status. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: event end to owner review deadline. Minimum evidence: signed/owner-issued completion records for required services and current unresolved-issue status. Method: apply a handover checklist; separate complete, evidence-missing, owner-deferred and not-applicable items; use telemetry only to target checks. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not declare the space clean, safe, undamaged or ready, and do not accept handover on an owner's behalf. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: show what is ready for owner review and what still needs evidence or inspection.

5. Planned and reactive maintenance

Prepare evidence, prioritise competent inspection and verify handback without issuing work or diagnosing beyond the measured service path.

FM-041 COMBINED MAINTENANCE WINDOW

L3 recommendation | R2 verified integration

FACILITIES MANAGERS QUESTION

Which approved maintenance tasks can share one window without increasing service or safety risk?

SENSORS AND TELEMETRY

Owner-data led; optional proposed telemetry. Scope: approved tasks, named assets, service dependencies and affected rooms/routes. Evidence: optional run state and demand context for affected services; the core answer is owner-controlled task/dependency data. Timing: current status plus forecast demand across candidate windows. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: CMMS owner, asset/service owners, permit/isolation owners, room/timetable owner, access/security and competent safety owners. Records: task scope, trade/competence, duration, dependencies, isolation, service impact, permits, access and approved windows. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: recommendation. Interval: candidate planning windows. Minimum evidence: current approved tasks with verified dependency, permission and impact data. Method: solve a constrained grouping problem; exclude incompatible isolations, simultaneous loss of resilience, access conflicts and owner restrictions. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not schedule work, authorise isolation, issue permits or certify risk; unresolved owner constraints block recommendation. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare the lowest-disruption combined-window proposal for owner approval.

FM-042 REACTIVE FAULT PRIORITY

L2 calculation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Which reactive fault warrants the earliest competent inspection?

SENSORS AND TELEMETRY

Proposed. Scope: named faulted assets, verified dependencies and affected services. Evidence: fault/alarm state, run/feedback, service-output indicators and point health where authorised. Timing: source-native events and 1-minute summaries from onset to current retrieval. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: CMMS owner, asset/service owner, BMS/specialist owner, safety/security owners and duty handover owner. Records: criticality, consequence, current owner status, affected service, contingency, access and open work. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: current reactive-response window. Minimum evidence: owner-confirmed fault or qualified abnormal condition, asset mapping and consequence evidence. Method: score consequence, persistence, service reach, contingency and evidence confidence; separate certified alarms, owner faults and telemetry anomalies. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not acknowledge alarms, dispatch work, certify safety or infer people affected; critical unknowns go to competent owners. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: set an inspection/escalation order with explicit evidence confidence.

5. Planned and reactive maintenance

Prepare evidence, prioritise competent inspection and verify handback without issuing work or diagnosing beyond the measured service path.

FM-043 LOW-DISRUPTION WORK PERIOD

L3 forecast | R2 verified integration

FACILITIES MANAGERS QUESTION

What authorised work window is likely to cause the least disruption for this noisy, dusty or service-interrupting task?

SENSORS AND TELEMETRY

Proposed. Scope: approved work area, affected service path, nearby eligible spaces and access route. Evidence: anonymous aggregate demand, numerical acoustic/environmental background and asset run state only where proportionate. Timing: 5-15 minute aggregates over comparable schedule windows plus forecast weather/event context. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: work owner, room/timetable/booking owners, access/security, cleaning, accessibility and facilities duty owners. Records: approved method, duration, noise/dust/service impact, room schedules, events, closures, access and contingency. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: forecast. Interval: candidate authorised windows. Minimum evidence: current owner constraints plus comparable demand evidence or an explicitly low-confidence fallback. Method: forecast aggregate exposure and service conflict for each eligible window; exclude prohibited periods and show prediction intervals. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not choose the final work time, infer individual schedules or treat low occupancy as access/permission. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: offer owners a ranked set of eligible low-disruption windows.

FM-044 POST-REPAIR SERVICE VERIFICATION

L2 comparison | R2 verified integration

FACILITIES MANAGERS QUESTION

After the repair, has the asset and affected service returned to its accepted operating pattern?

SENSORS AND TELEMETRY

Proposed. Scope: repaired asset, relevant service-output points and verified affected rooms/zones. Evidence: pre/post run state, command/feedback, load, temperature, pressure, flow, current or vibration features as applicable. Timing: source-native events and 1-minute summaries across accepted pre-fault, post-handback and current windows. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: CMMS owner, competent engineer/contractor handback owner, asset/BMS owner and service owner. Records: repair scope/time, acceptance criteria, commissioning results, current schedule/load, overrides and unresolved defects. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: post-handback stabilisation window versus accepted baseline. Minimum evidence: signed handback, same asset identity, comparable operation and healthy points. Method: compare accepted service indicators and variability; separate restoration, partial recovery, new abnormality and sensor uncertainty. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not accept handback, close the work order or certify safety; unmatched conditions yield a qualified result. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare evidence for the competent owner's handback or follow-up decision.

5. Planned and reactive maintenance

Prepare evidence, prioritise competent inspection and verify handback without issuing work or diagnosing beyond the measured service path.

FM-045 EARLY DEGRADATION SIGNAL

L3 forecast | R2 verified integration

FACILITIES MANAGERS QUESTION

Which maintainable assets show repeatable deterioration early enough to justify a condition check before failure?

SENSORS AND TELEMETRY

Proposed. Scope: maintainable rotating/electrical/mechanical assets with stable point mapping and service context. Evidence: vibration features, current, temperature, differential pressure, flow, run hours, starts and fault history as applicable. Timing: source-native events plus owner-approved feature summaries over sufficient operating cycles. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: asset owner, condition-monitoring owner, CMMS owner, BMS/specialist owner and maintenance-history owner. Records: asset model/age, expected operating envelope, maintenance, failures, load/regime and sensor/firmware changes. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: forecast. Interval: owner-defined planning horizon. Minimum evidence: stable asset identity, enough valid operating cycles, current maintenance state and an accepted condition baseline. Method: model trend and remaining margin with confidence bounds; test sensitivity to load/regime and sensor drift; flag only repeatable deterioration. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not predict an exact failure date, diagnose internally or create work automatically. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: select assets for competent condition inspection before a service loss.

FM-046 SENSOR OR PHYSICAL FAULT

L3 diagnosis | R2 verified integration

FACILITIES MANAGERS QUESTION

Is this abnormal reading more consistent with a sensing problem or a real asset or room condition?

SENSORS AND TELEMETRY

Proposed. Scope: suspect point, measured asset/room, independent reference or redundant point and relevant service context. Evidence: suspect value, point-health diagnostics, reference points, command/feedback and physical service indicators. Timing: native cadence over onset and recovery plus a recent healthy baseline. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: sensor/BMS owner, asset owner, calibration/commissioning owner, CMMS owner and room/service owner. Records: point mapping, placement, calibration, maintenance/firmware, known faults, asset state and recent work. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: diagnosis. Interval: anomaly window plus healthy baseline. Minimum evidence: current mapping, suspect and independent evidence, aligned clocks and known maintenance state. Method: test plausibility, redundancy, physical consistency and response to known state changes; rank sensor, mapping and physical explanations. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not suppress alarms, recalibrate remotely or conclude the physical condition is safe because a sensor seems faulty. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: choose sensor validation, physical inspection or owner escalation as the next step.

5. Planned and reactive maintenance

Prepare evidence, prioritise competent inspection and verify handback without issuing work or diagnosing beyond the measured service path.

FM-047 OUTAGE IMPACT SCOPE

L2 calculation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

If this asset is taken offline, which rooms and services fall within its verified impact boundary?

SENSORS AND TELEMETRY

Proposed. Scope: named asset, upstream/downstream topology, served rooms/zones and authorised contingency paths. Evidence: current run state and redundant-path status only as context; topology and owner records are primary. Timing: current status at retrieval and proposed outage window. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: asset/service topology owner, room/service owners, CMMS owner, permit/isolation owner, security/safety owners and contingency owner. Records: current dependency graph, isolation plan, redundancy, approved outage, room/service criticality and current closures. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: proposed outage window. Minimum evidence: verified current topology and owner-approved isolation/contingency records. Method: traverse dependencies and classify direct, dependent, redundant and unknown impacts; expose stale or conflicting topology. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not authorise the outage, isolation or access, and do not infer occupant locations or safety. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare an impact list for competent-owner outage planning and communication.

FM-048 ENGINEER EVIDENCE PACK

L2 calculation | R2 verified integration

FACILITIES MANAGERS QUESTION

What current, traceable evidence should I prepare for the competent engineer investigating this fault?

SENSORS AND TELEMETRY

Proposed. Scope: named asset, relevant upstream/downstream service points and affected spaces. Evidence: only the minimum fault events, operating trends, point-health and service-output evidence needed for the stated fault. Timing: owner-approved event window plus matched baseline at native or 1-minute resolution. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: CMMS owner, asset/BMS owner, competent engineer, document/permit owner and room/service owner. Records: asset identity/topology, fault history, maintenance, drawings/configuration, warranties, permits, overrides and recent changes. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: engineer's declared diagnostic window. Minimum evidence: verified asset identity, fault scope and minimum requested evidence with permission. Method: assemble a provenance-preserving pack; align clocks, flag gaps/drift and keep raw observation separate from interpretation. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not expose credentials/restricted security data, diagnose for the engineer or authorise work. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: reduce duplicate data gathering and give the engineer a defensible starting record.

5. Planned and reactive maintenance

Prepare evidence, prioritise competent inspection and verify handback without issuing work or diagnosing beyond the measured service path.

FM-049 REPEAT CALLOUT ROOT CAUSE

L3 diagnosis | R2 verified integration

FACILITIES MANAGERS QUESTION

Do repeated callouts justify widening the investigation beyond another local repair?

SENSORS AND TELEMETRY

Proposed. Scope: repeatedly affected asset, local components, upstream dependencies and served spaces. Evidence: fault events, run/load/service-output trends, condition features and point health across each callout. Timing: aligned event windows for all callouts plus matched normal operation. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: CMMS owner, asset/service owners, competent engineer/contractor records and change/maintenance owners. Records: callout cause/action, parts, handback, recurrence, asset topology, operating regime and known design constraints. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: diagnosis. Interval: multi-callout history and current issue window. Minimum evidence: at least two comparable callouts with traceable actions and valid evidence. Method: compare recurrence signatures and post-work persistence; rank local repair, upstream cause, design/operation issue and evidence gap. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not blame a contractor, prove causality or approve redesign from correlation alone. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: decide whether to escalate to broader competent investigation or continue local maintenance.

FM-050 BLOCKED MAINTENANCE DEPENDENCIES

L2 calculation | R1 owner-data ready

FACILITIES MANAGERS QUESTION

Which approved maintenance tasks cannot proceed because a part, permit, access, isolation or owner decision is missing?

SENSORS AND TELEMETRY

Owner-data led; optional proposed telemetry. Scope: approved tasks and their named assets/work areas; telemetry is optional status context only. Evidence: optional asset availability or isolation-status heartbeat where owner-approved. Timing: current authoritative state at retrieval and proposed work window. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: CMMS owner, stores/procurement, permit/isolation owners, access/security, room/service owners and contractor coordinator. Records: task approval, part availability, competence, permit, RAMS, access, isolation, outage, dependencies and owner decisions. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: current planning cycle. Minimum evidence: current approved tasks and mandatory prerequisite checklist. Method: evaluate prerequisites and classify ready, blocked, conflict or owner-deferred; name exactly one next owner action for each block. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not issue permits, grant access, order parts, schedule work or infer approval from telemetry. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: focus coordination on the dependency that actually prevents progress.

6. Contractors, permits, access and handovers

Check prerequisites, impact boundaries and evidence completeness while permit, access and competent-owner decisions remain authoritative.

FM-051 CONTRACTOR VISIT PREREQUISITES

L2 calculation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Which contractor visits scheduled today have complete prerequisites, and which need owner clarification before arrival?

SENSORS AND TELEMETRY

Owner-data led; optional proposed telemetry. Scope: scheduled work areas and named assets; optional owner-issued asset/area status only. Evidence: optional door/asset status heartbeat; no tracking of contractors or personal devices. Timing: current authoritative state at retrieval and visit window. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: contractor coordinator, CMMS owner, permit/RAMS owner, access/security, asset/room owners and facilities duty owner. Records: approved visit, competence, contact, work scope, RAMS, permit, access, induction, isolation, parts and area status. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: today's visit windows. Minimum evidence: current scheduled visits and complete mandatory prerequisite records. Method: apply a prerequisite checklist; classify ready for owner review, clarification needed, cancelled/changed by owner or not assessable. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not identify/track contractors, grant access, approve RAMS/permits or confirm attendance. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: contact the right owner before arrival and avoid preventable failed visits.

FM-052 CONTRACTOR WORK IMPACT

L2 calculation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Which rooms, routes and services are within the verified impact scope of this approved contractor work?

SENSORS AND TELEMETRY

Owner-data led; optional proposed telemetry. Scope: approved work boundary, named assets, service topology, access route and temporary isolations/diversions. Evidence: current owner-issued isolation/diversion state and optional service-status telemetry. Timing: current status plus approved work window. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: work owner, asset/service topology owner, room/route owners, permit/isolation owner, security/access, accessibility and safety owners. Records: approved scope, RAMS, permit, isolation, route, service impact, contingency, communications and handover plan. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: approved work window. Minimum evidence: current approved work documents and verified topology/route mapping. Method: calculate direct, dependent and temporary impacts; retain owner-declared and telemetry-observed status separately; show unknown edges. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not expand the work scope, grant access, authorise isolation or disclose restricted security details. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare accurate facilities coordination and affected-service communication.

6. Contractors, permits, access and handovers

Check prerequisites, impact boundaries and evidence completeness while permit, access and competent-owner decisions remain authoritative.

FM-053 WORK-AREA HANDOVER READINESS

L2 calculation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Is the recorded evidence complete enough for the authorised owner to review handover of this work area?

SENSORS AND TELEMETRY

Owner-data led; optional proposed telemetry. Scope: approved work area, affected assets/services and required restoration points. Evidence: owner-required service/asset status and optional room environmental or utility baseline evidence. Timing: post-work period through declared handover time. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: work/permit owner, contractor coordinator, asset/room owners, safety/security owners and facilities duty owner. Records: completion record, inspections/tests, restored isolations/barriers, defects, waste clearance, keys/access and outstanding actions. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: declared handover review window. Minimum evidence: owner-required handover checklist and signed source records. Method: classify each prerequisite as complete, conflicting, missing or not applicable; use telemetry only to target verification. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not accept handover, certify safety/compliance, track contractors or infer work quality from absence of alarms. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: show the authorised owner exactly what is ready for review and what blocks handover.

FM-054 PERMIT EXPIRY RISK

L2 estimate | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Which permits or temporary approvals may expire before work and restoration are recorded complete?

SENSORS AND TELEMETRY

Owner-data led; optional proposed telemetry. Scope: approved work areas/assets; sensing is not required for the core answer. Evidence: optional owner-issued work-state and isolation/barrier status only. Timing: current record state at retrieval and forecast to planned completion/restoration. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: permit-to-work owner, work owner, contractor coordinator, access/security and asset/service owners. Records: permit issue/expiry, approved extensions, work programme, restoration dependencies, handover state and conflicts with RAMS/site status. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: estimate. Interval: current time to planned completion and restoration. Minimum evidence: current permit/approval records and an owner-approved programme. Method: calculate time margin and dependency risk; show record conflicts and uncertainty rather than predicting worker behaviour. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not extend/approve a permit, infer work progress from occupancy or advise work to continue after expiry. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: escalate expiring authority to the permit owner before it becomes a stop-work issue.

6. Contractors, permits, access and handovers

Check prerequisites, impact boundaries and evidence completeness while permit, access and competent-owner decisions remain authoritative.

FM-055 TEMPORARY ISOLATION AND BARRIER STATUS

L2 comparison | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Are temporary isolations, barriers and service diversions still within their authorised scope and time?

SENSORS AND TELEMETRY

Proposed. Scope: owner-declared isolation/barrier/diversion boundary and affected assets/routes. Evidence: owner-issued status; contact/position sensing only where part of an authorised safety system. Timing: event-driven owner status and current retrieval against approved start/expiry. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: permit/isolation owner, security/access, route/accessibility owner, asset/service owner and facilities duty owner. Records: approved scope/time, inspection frequency, diversion route, extension, restoration and current owner acknowledgement. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: current time within the authorised work window. Minimum evidence: current approved records and authorised status source. Method: compare current status with approved scope/time; classify confirmed, stale, conflicting or unknown; no inference from silence. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not remove barriers, restore services, certify egress/accessibility or override permit/security owners. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: identify the exact owner confirmation needed before the temporary arrangement continues.

FM-056 PRE-WORK EVIDENCE CAPTURE

L2 calculation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

What minimum condition evidence should be captured before this contractor starts work?

SENSORS AND TELEMETRY

Proposed. Scope: approved work asset/area and owner-defined adjacent assets or spaces at risk. Evidence: only relevant asset state, meter/utility baseline, numerical environmental condition and point health; no content capture. Timing: owner-approved pre-work window at source-native or 1-minute resolution. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: work owner, competent engineer, permit/RAMS owner, asset/room owners, data owners and facilities duty owner. Records: approved scope, risk controls, acceptance/handback criteria, required photographs/documents by authorised staff and evidence retention rules. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: declared pre-work baseline window. Minimum evidence: approved work scope and owner-defined baseline checklist with permission. Method: select the minimum evidence needed to distinguish pre-existing condition from work effects; preserve provenance and gaps. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not capture people, speech, screens or restricted content, and do not authorise work if prerequisites are missing. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: create a defensible baseline for later handback and dispute avoidance.

6. Contractors, permits, access and handovers

Check prerequisites, impact boundaries and evidence completeness while permit, access and competent-owner decisions remain authoritative.

FM-057 POST-WORK HANDBACK

L2 comparison | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Are the owner-required restoration checks complete for post-work handback review?

SENSORS AND TELEMETRY

Proposed. Scope: worked asset/area, restored service points, isolations/barriers and affected rooms/routes. Evidence: post-work asset/service status and the same approved baseline points used before work. Timing: source-native events and 1-minute summaries from completion through stabilisation. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: work/permit owner, contractor coordinator, asset/room owners, competent engineer and facilities duty owner. Records: contractor completion, tests, defects, restored configuration, permits/isolations, cleaning/waste, keys and owner acceptance criteria. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: completion through owner review. Minimum evidence: signed completion evidence plus comparable pre-work baseline and valid post-work points. Method: compare current state with owner acceptance criteria and baseline; keep contractor assertions, facilities observations and telemetry distinct. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not accept handback, close permits, certify safety or infer contractor activity. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare a clear owner review of restored, unresolved and evidence-missing items.

FM-058 WORK AND EVENT CONFLICT

L2 comparison | R2 verified integration

FACILITIES MANAGERS QUESTION

Which planned contractor activities overlap with teaching, events or other critical room use?

SENSORS AND TELEMETRY

Owner-data led; optional proposed telemetry. Scope: approved work areas, affected service routes and current booked/event spaces. Evidence: optional anonymous aggregate demand forecast and asset status; schedule data is primary. Timing: current plans across the candidate work/event windows. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: work owner, room-booking/timetetable/event owners, room/service owners, access/security and accessibility owners. Records: approved work scope/time, service interruption, bookings/events, setup/handback, access routes and contingency. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: planning horizon for the work and event. Minimum evidence: current approved work and authoritative room/event schedules with verified impact mapping. Method: identify temporal, spatial, service and route conflicts; show whether alternative windows are authorised or merely possible. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not move/cancel events, reschedule work, grant access or infer attendance. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: coordinate an owner-led resolution before either activity is disrupted.

6. Contractors, permits, access and handovers

Check prerequisites, impact boundaries and evidence completeness while permit, access and competent-owner decisions remain authoritative.

FM-059 CONTRACTOR DEPENDENCY ESCALATION

L2 calculation | R1 owner-data ready

FACILITIES MANAGERS QUESTION

Which unresolved contractor dependencies need owner escalation before the scheduled arrival?

SENSORS AND TELEMETRY

Owner-data led; optional proposed telemetry. Scope: scheduled work packages and named prerequisites; no sensing is required for the core answer. Evidence: optional owner-issued asset availability or area-status context. Timing: current record state and time remaining to arrival. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: contractor coordinator, CMMS owner, permit/RAMS, stores, access/security, asset/room owners and finance/procurement where relevant. Records: approved scope, parts, documentation, competence, permit, access, isolation, keys, owner decisions and contact chain. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: current time to scheduled arrival. Minimum evidence: current scheduled visit and mandatory prerequisite checklist. Method: calculate readiness and time-to-resolution for each dependency; name the owner and smallest decision needed. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not contact personal data beyond authorised workflow, grant access, approve documents or infer contractor travel/attendance. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: escalate the blocker most likely to cause a failed or delayed visit.

FM-060 HANDOVER DISCREPANCIES

L3 diagnosis | R2 verified integration

FACILITIES MANAGERS QUESTION

Which differences remain between the contractor completion record and the current observed service state?

SENSORS AND TELEMETRY

Proposed. Scope: worked asset/area, declared restored configuration and verified affected services. Evidence: current asset/service state, point health and owner-approved post-work baseline evidence. Timing: source-native events and 1-minute summaries from declared completion to current retrieval. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: contractor/work owner, CMMS owner, asset/BMS owner, permit owner and competent engineer. Records: completion statement, test results, configuration, defects, handback criteria, current owner status and recent changes. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: diagnosis. Interval: post-completion stabilisation and current owner review window. Minimum evidence: traceable completion record plus current valid observations and mapping. Method: compare declared and observed state; rank stale record, incomplete restoration, new fault, configuration mismatch and sensor issue. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not accuse a contractor, prove causality, accept/reject handover or alter configuration. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: direct the authorised owner to the smallest discrepancy requiring verification.

7. Incidents, closures, security and emergency coordination

Support facilities coordination from certified and owner-issued sources without replacing incident command, safety or security authority.

FM-061 INCIDENT OWNERSHIP AND FACILITIES ROLE

L2 calculation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

For each current incident, what facilities coordination is required and which decisions remain with another competent owner?

SENSORS AND TELEMETRY

Proposed. Scope: owner-declared incident zones, affected facilities assets/services and authorised coordination interfaces. Evidence: certified/owner status, utility/plant state and only incident-approved telemetry; no person tracking. Timing: source-native certified events and current owner updates across the incident period. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: incident commander, security, fire/health-and-safety, certified-system owners, facilities duty owner and asset/service owners. Records: current incident classification, command structure, declared zones/routes, closures, requested facilities tasks and communication rules. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: current incident command period. Minimum evidence: current authoritative incident record and named facilities task/owner. Method: map each requested action to responsible owner, prerequisite and evidence; label telemetry context separately from command status. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not issue emergency instructions, acknowledge alarms, grant access, operate systems or replace command decisions. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: keep facilities coordination inside its authorised role and avoid conflicting actions.

FM-062 CURRENT CLOSURE IMPACT

L2 calculation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

What spaces are officially closed now, and which facilities services or authorised routes fall within scope?

SENSORS AND TELEMETRY

Proposed. Scope: owner-declared closure boundary, affected rooms/routes/assets and approved alternatives. Evidence: door/lift/service status only as authorised context; no occupancy inference. Timing: current owner status at retrieval and closure effective period. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: closure owner, room/route owners, security/access, accessibility, facilities service owners and communications owner. Records: official closure, effective/expiry time, affected services/routes, exceptions, approved alternatives and current conflicts. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: current closure window. Minimum evidence: authoritative closure record and verified current topology/route mapping. Method: traverse declared closure impacts and separate direct, dependent, alternative and unknown services/routes. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not infer occupant locations, open/close spaces, grant access or certify alternative-route accessibility/safety. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: coordinate facilities support and communicate only verified closure effects.

7. Incidents, closures, security and emergency coordination

Support facilities coordination from certified and owner-issued sources without replacing incident command, safety or security authority.

FM-063 LEAK OR POWER INCIDENT SPREAD

L3 diagnosis | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Does verified evidence show the reported leak or electrical issue extending beyond the initially declared area?

SENSORS AND TELEMETRY

Proposed. Scope: declared incident area, upstream/downstream water/electrical paths and adjacent services within authorised scope. Evidence: leak sensors, water flow/pressure, breaker/protection status, voltage/current/power quality and owner-issued incident state. Timing: source-native events and 1-minute trends from onset to current retrieval. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: incident commander, electrical/water competent owners, certified-system owners, security/safety and facilities duty owner. Records: declared zone, isolation status, asset topology, reports, current hazards, maintenance and incident chronology. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: diagnosis. Interval: incident onset to current command update. Minimum evidence: authoritative incident boundary plus valid mapped service evidence and aligned clocks. Method: compare event order and network/branch propagation; classify confirmed, possible, conflicting and unknown spread. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not declare the area safe, energise/isolate systems or override incident command; indirect evidence cannot expand the official zone. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: give competent owners evidence for targeted inspection and boundary review.

FM-064 AUTHORISED FACILITIES ACCESS ROUTE

L2 calculation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Which current route is authorised for facilities attendance during this closure or incident?

SENSORS AND TELEMETRY

Proposed. Scope: authorised incident routes, doors, lifts, access points and declared exclusion zones. Evidence: owner-issued door/lift/route status; anonymous aggregate congestion only if incident authority permits. Timing: current status with retrieval time and owner-defined expiry/refresh interval. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: incident commander, security/access owner, fire/safety, route/accessibility owner and facilities duty owner. Records: authorised route, access permission, escort/PPE requirements, exclusions, lift/door status and contingency. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: immediate attendance window. Minimum evidence: current command-approved route and access conditions. Method: validate the whole route against current authoritative states; present alternatives only if also approved; show stale segments as unknown. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not grant access, reveal sensitive routes, certify personal safety or operate doors/lifts. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: select a currently authorised route or defer attendance until the responsible authority confirms one.

7. Incidents, closures, security and emergency coordination

Support facilities coordination from certified and owner-issued sources without replacing incident command, safety or security authority.

FM-065 POST-INCIDENT EVIDENCE TIMELINE

L2 calculation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Which facilities records and eligible telemetry can support the competent authority's post-incident timeline?

SENSORS AND TELEMETRY

Proposed. Scope: declared incident assets/services and only data sources approved for post-incident use. Evidence: certified events, owner logs, utility/plant state, selected environmental/asset telemetry and point health. Timing: source-native timestamps with documented clock offsets across the declared incident window. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: incident owner, certified-system owners, security/safety, CMMS/BMS/utility owners and records/governance owner. Records: official chronology, incident log, alarms, work/closure actions, communications, sensor mapping/health and retention/permission rules. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: declared incident window plus recovery period. Minimum evidence: authorised source list, documented clocks and traceable provenance. Method: normalise timestamps without altering originals; build an evidence timeline and mark conflicts, gaps and inferred links explicitly. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not infer people, responsibility or causality, expose restricted information or replace formal investigation. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare a traceable facilities evidence package for competent review.

FM-066 CERTIFIED STATUS BLIND SPOTS

L2 calculation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Which certified alarm or safety-system statuses are unavailable, stale or conflicting in the facilities view?

SENSORS AND TELEMETRY

Proposed. Scope: authorised certified-system interfaces and their communications/heartbeat path. Evidence: owner-issued system state, interface heartbeat and last successful update only; no substitute IoT alarm. Timing: event-driven source status with owner-defined freshness thresholds. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: certified-system owner, fire/health-and-safety, security, incident command and facilities duty owner. Records: system/interface inventory, current certified state, maintenance/test status, communication path, permissions and known outages. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: current operational or incident period. Minimum evidence: authoritative system inventory and owner-defined freshness criteria. Method: classify each interface as current, stale, conflicting, unavailable or out of facilities scope; do not infer state from silence. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not replace certified alarms, acknowledge events, declare safety or expose restricted system detail. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: escalate the exact visibility gap to the certified-system owner.

7. Incidents, closures, security and emergency coordination

Support facilities coordination from certified and owner-issued sources without replacing incident command, safety or security authority.

FM-067 POST-EVENT ASSET INSPECTION

L3 recommendation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

After the authoritative event, which facilities assets meet the owner-defined criteria for competent inspection?

SENSORS AND TELEMETRY

Proposed. Scope: assets within the declared event/incident impact boundary and their verified dependencies. Evidence: event-approved condition indicators, fault events, run state and point health only after authority permits access. Timing: event window through post-clearance observation period. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: incident/event owner, asset/service owners, CMMS owner, competent engineer and facilities duty owner. Records: official event scope, clearance, inspection criteria, affected assets, prior condition, maintenance and current access restrictions. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: recommendation. Interval: post-clearance inspection-planning window. Minimum evidence: competent clearance plus valid asset mapping and owner-defined trigger criteria. Method: rank assets by confirmed exposure, abnormal condition, dependency and evidence confidence; keep unknown interfaces unknown. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not enter before clearance, certify safety, dispatch work automatically or treat silence as healthy. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare a competent inspection sequence after authority release.

FM-068 SERVICE CONTINGENCY DURING OUTAGE

L2 comparison | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

During this lift, utility or plant outage, which owner-approved facilities contingency is currently available?

SENSORS AND TELEMETRY

Proposed. Scope: outage asset/service, approved backup or alternative service path and affected rooms/routes. Evidence: owner-issued primary/backup status and relevant capacity/health points where authorised. Timing: current status with owner-defined heartbeat and outage window. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: outage owner, contingency owner, room/route/accessibility owners, security/safety and facilities duty owner. Records: official outage, approved contingency, capacity/limits, access, duration, dependencies and current conflicts. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: current outage period. Minimum evidence: current owner-approved contingency and valid status for all required components. Method: compare primary loss and approved alternative state; show capacity/coverage limits and chronology by source. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not activate contingency, guarantee capacity/accessibility or invent an alternative outside owner plans. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: communicate the available approved option or defer when status is incomplete.

7. Incidents, closures, security and emergency coordination

Support facilities coordination from certified and owner-issued sources without replacing incident command, safety or security authority.

FM-069 PLANNED EMERGENCY TEST IMPACT

L2 calculation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

Which facilities services and scheduled activities fall within the declared impact of the authorised emergency-system test?

SENSORS AND TELEMETRY

Owner-data led; optional proposed telemetry. Scope: declared test zones/systems, affected facilities assets, rooms, routes and approved exemptions. Evidence: certified test status and owner-issued affected-service state; optional facilities telemetry only as context. Timing: official test window and source-native events. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: test authority, certified-system owner, fire/health-and-safety, security, room/timetable/event owners and facilities duty owner. Records: test plan, affected zones/services, notifications, access, exemptions, restoration and clearance requirements. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: authorised test window through declared restoration. Minimum evidence: current approved test plan and verified impact mapping. Method: map direct and dependent facilities effects; separate planned test signals from genuine faults only as owner defines. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not silence/acknowledge alarms, cancel activities, grant clearance or interpret certified signals independently. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare facilities coordination and follow-up checks around the test.

FM-070 RETURN TO NORMAL SERVICE

L2 calculation | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

After competent-authority clearance, what facilities evidence is still outstanding before normal support resumes?

SENSORS AND TELEMETRY

Proposed. Scope: cleared incident/closure scope, affected assets/services and owner-defined restoration points. Evidence: owner-issued restoration state plus approved asset/service status evidence. Timing: clearance time through current recovery window. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: incident/closure authority, asset/service owners, CMMS owner, security/access, room/route owners and facilities duty owner. Records: formal clearance, restoration tests, open defects, temporary controls, access/routes, communications and owner acceptance. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: post-clearance recovery period. Minimum evidence: formal clearance plus owner-defined restoration checklist. Method: classify restored, restricted, temporary, evidence-missing and owner-deferred services; preserve source chronology and conflicts. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not reopen spaces, certify safety, remove controls or declare normal service on behalf of competent owners. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: coordinate the remaining facilities checks and communicate qualified service status.

8. Performance, lifecycle, resilience and sustainability

Compare delivered service, condition, cost and resource evidence for planning while keeping estimates separate from approval or compliance.

FM-071 ENERGY PER DELIVERED SERVICE

L2 calculation | R2 verified integration

FACILITIES MANAGERS QUESTION

Which assets use unusually high energy relative to their measured delivered service under comparable operation?

SENSORS AND TELEMETRY

Proposed. Scope: stable meter boundary, named asset/system and measured delivered-service output. Evidence: electricity/gas/heat meter data, run hours, flow/pressure/thermal output and load/weather context. Timing: 1-minute or 15-minute owner-approved aggregates over representative operating periods. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

ANALYSIS REQUIRED

Operation: calculation. Interval: representative reporting period and matched operating regimes. Minimum evidence: stable meter and service-output boundaries, aligned clocks and adequate load/weather coverage. Method: calculate energy per service unit; normalise by load/regime and compare eligible peers or baseline; show measurement and model uncertainty. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

AUTHORITATIVE SUPPORTING DATA

Owners: meter/utility owner, asset/BMS owner, service owner, room/use owner, weather source and finance/carbon-factor owner. Records: meter hierarchy, tariff/factors, delivered-service definition, schedule, maintenance, overrides and known boundary changes. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not present the result as audited savings, compliance or proof of waste, and do not compare mismatched boundaries. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: identify assets for engineering review or better metering before investment decisions.

FM-072 FAULT CONSEQUENCE AND COST

L2 calculation | R2 verified integration

FACILITIES MANAGERS QUESTION

Which recurring faults create the greatest combined service disruption, callout effort and recorded cost?

SENSORS AND TELEMETRY

Proposed. Scope: faulted assets, verified affected services and only telemetry needed to validate duration/state. Evidence: fault events, service-output loss and operating context where available. Timing: source-native events across the selected lifecycle/reporting period. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

ANALYSIS REQUIRED

Operation: calculation. Interval: selected annual or lifecycle reporting period. Minimum evidence: traceable fault/cost records, stable asset identity and explicit consequence definition. Method: deduplicate events; calculate frequency, duration, service consequence and recorded cost; normalise for operating exposure and show missing-cost sensitivity. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

AUTHORITATIVE SUPPORTING DATA

Owners: CMMS/finance owners, asset/service owners, contractor/callout records and room/event owners. Records: fault classifications, downtime, affected service, callout labour/parts, consequential cost rules and data-quality limits. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not rank staff/contractors, infer blame or treat unrecorded consequences as zero. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: focus lifecycle review on faults with the strongest documented combined consequence.

8. Performance, lifecycle, resilience and sustainability

Compare delivered service, condition, cost and resource evidence for planning while keeping estimates separate from approval or compliance.

FM-073 CONDITIONING OR LIGHTING UNUSED SPACE

L2 calculation | R2 verified integration

FACILITIES MANAGERS QUESTION

Where are heating, cooling, ventilation or lighting operating while authoritative records show no planned use?

SENSORS AND TELEMETRY

Proposed. Scope: rooms/zones with current purpose, schedule, access and verified BMS/lighting mapping. Evidence: anonymous occupancy aggregate, HVAC/lighting run state, power/flow and environmental context. Timing: 5-minute aggregates across matched planned-unoccupied periods. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: room/timetable/booking owners, BMS/lighting owners, access/opening owner and facilities duty owner. Records: authoritative planned use, cancellations, approved exceptions, cleaning/maintenance, setpoints, overrides and closures. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: selected planned-unoccupied windows. Minimum evidence: current schedule/exception data, verified mapping and valid operation state. Method: calculate duration and estimated resource use during planned vacancy; separate authoritative vacancy from uncertain occupancy and authorised service needs. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not infer individual presence, switch systems off or call any use waste without owner context. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: identify schedule, control or exception records for owner review and quantify only defensible opportunity.

FM-074 SUSTAINABILITY INTERVENTION OUTCOME

L2 comparison | R2 verified integration

FACILITIES MANAGERS QUESTION

Has the intervention reduced energy or resource use while preserving the measured service outcome under comparable conditions?

SENSORS AND TELEMETRY

Proposed. Scope: intervention assets/zones, stable meter boundary and service/environment outcome points. Evidence: energy/water data, delivered service, room environmental indicators, aggregate demand and weather/load context. Timing: owner-approved aggregation across matched pre- and post-intervention periods. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: intervention owner, meter/utility owner, asset/BMS owner, room/use owner, finance/carbon-factor owner and CMMS owner. Records: approved intervention/time, intended outcome, cost, maintenance, confounders, current factors and acceptance criteria. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: matched pre/post periods after stabilisation. Minimum evidence: traceable intervention, stable boundaries and adequate comparable operation before/after. Method: normalise for weather/load/use; compare resource and service outcomes; report confidence, spillovers and sensitivity to missing data. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not claim causality, audited savings, carbon certification or user benefit beyond measured outcomes. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: decide whether evidence supports continuation, further monitoring or competent review.

8. Performance, lifecycle, resilience and sustainability

Compare delivered service, condition, cost and resource evidence for planning while keeping estimates separate from approval or compliance.

FM-075 END-OF-LIFE CANDIDATES

L3 forecast | R2 verified integration

FACILITIES MANAGERS QUESTION

Which assets have enough condition, reliability and cost evidence to justify replacement-planning review?

SENSORS AND TELEMETRY

Proposed. Scope: maintainable assets with stable identity, condition/service evidence and comparable lifecycle records. Evidence: condition features, run hours, faults, service output and energy where appropriate. Timing: owner-approved aggregates across sufficient operating cycles and lifecycle period. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: asset/lifecycle owner, CMMS/finance owners, BMS/condition-monitoring owner and service owner. Records: age/model, criticality, failures, repairs, downtime, parts obsolescence, condition, energy, replacement constraints and cost rules. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: forecast. Interval: owner-defined capital-planning horizon. Minimum evidence: stable asset history, traceable costs and enough valid condition/reliability evidence. Method: estimate degradation and expected maintenance/service burden with uncertainty; compare eligible peers or manufacturer/owner baseline. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not approve capital spend, predict exact failure or present association as guaranteed savings. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prepare a defensible shortlist for asset-owner replacement planning.

FM-076 SENSING GAP PRIORITY

L3 recommendation | R2 verified integration

FACILITIES MANAGERS QUESTION

Where would added, relocated or higher-quality sensing most reduce facilities decision risk?

SENSORS AND TELEMETRY

Proposed. Scope: catalogue decisions, their required assets/rooms/services and current point coverage. Evidence: inventory, mapping, cadence, quality, calibration/functional checks, uptime/gaps, drift, redundancy and decision dependencies. Timing: quality assessed over the actual decision intervals, not only current heartbeat. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: sensor/IoT owner, BMS/specialist owners, asset/room owners, CMMS owner and data-governance owner. Records: question-to-evidence requirements, current inventory/health, consequence, installation constraints, permissions and maintenance burden. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: recommendation. Interval: next sensing/commissioning planning cycle. Minimum evidence: explicit decision requirements, current coverage/quality and consequence of error. Method: score each gap by decision consequence, frequency, uncertainty reduction, installation feasibility and lifecycle cost; include no-sensor alternatives. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not assume more sensing is always better, propose intrusive monitoring or treat historical-plan locations as current. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prioritise the smallest sensing changes that materially improve facilities decisions.

8. Performance, lifecycle, resilience and sustainability

Compare delivered service, condition, cost and resource evidence for planning while keeping estimates separate from approval or compliance.

FM-077 SERVICE RESILIENCE TO SINGLE FAILURE

L2 comparison | R3 restricted or higher-consequence

FACILITIES MANAGERS QUESTION

How does the documented service design respond to loss of one critical asset or utility path?

SENSORS AND TELEMETRY

Proposed. Scope: critical service topology, approved redundancy/backup paths and tested status interfaces. Evidence: owner-issued primary/backup state, capacity and test result where authorised; no active failure injection. Timing: current status plus owner-approved historical test/incident evidence. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: critical-service owner, asset/topology owner, safety/security, incident/business-continuity owner and facilities duty owner. Records: current design/topology, redundancy limits, capacity, dependencies, last test, failures, maintenance and approved contingency. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: declared resilience scenario and planning horizon. Minimum evidence: verified topology and owner-approved contingency/test evidence. Method: compare normal and single-loss service paths; identify direct loss, reduced capacity, manual dependency and unknown edge; state test freshness. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not simulate by operating systems, disclose sensitive topology, certify resilience or override contingency owners. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: identify untested dependencies and owner actions for resilience planning.

FM-078 SEASONAL OPERATING PRIORITIES

L3 recommendation | R2 verified integration

FACILITIES MANAGERS QUESTION

Before the next academic period, which seasonal operating checks should facilities schedule first?

SENSORS AND TELEMETRY

Proposed. Scope: season-sensitive plant, served zones, public-facing services and relevant monitoring points. Evidence: condition/run state, room environment, utility demand, weather exposure and point health as appropriate. Timing: owner-approved summaries across the prior comparable season and current pre-season state. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: seasonal plan owner, asset/BMS owners, room/timetable/event owners, utility/weather sources, CMMS and service owners. Records: academic calendar, seasonal procedures, known faults, prior complaints/incidents, maintenance, closures and contingency. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: recommendation. Interval: pre-season planning window. Minimum evidence: current owner plans, verified topology and representative prior-season evidence. Method: risk-rank checks by service consequence, unresolved defects, prior seasonal performance, evidence gaps and limited contingency. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not start plant, authorise work, infer individual behaviour or certify readiness. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: build a focused pre-season verification programme for owner approval.

8. Performance, lifecycle, resilience and sustainability

Compare delivered service, condition, cost and resource evidence for planning while keeping estimates separate from approval or compliance.

FM-079 REPAIR OR REPLACEMENT EVIDENCE

L2 comparison | R2 verified integration

FACILITIES MANAGERS QUESTION

Does the current evidence favour another repair or preparation of a replacement business case for this asset?

SENSORS AND TELEMETRY

Proposed. Scope: named asset, verified service output, condition and stable meter/cost boundaries. Evidence: condition indicators, run hours, faults, service performance and energy where available. Timing: owner-approved aggregates across the relevant lifecycle and recent fault period. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: asset owner, CMMS/finance/procurement owners, competent engineer, BMS/condition owner and service owner. Records: repair history/cost, downtime, parts, condition, criticality, obsolescence, replacement constraints and accepted financial rules. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: comparison. Interval: owner-defined repair/replacement planning horizon. Minimum evidence: traceable repair/cost history, current condition/service evidence and explicit decision criteria. Method: compare expected repair path and replacement-case evidence using ranges and sensitivity; show missing lifecycle or benefit evidence. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not approve spend, promise savings, select a supplier or replace competent engineering judgement. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: decide whether to commission more evidence, attempt repair or develop a formal replacement case.

FM-080 UNANSWERABLE PERFORMANCE CLAIMS

L2 calculation | R1 owner-data ready

FACILITIES MANAGERS QUESTION

Which facilities performance questions cannot currently be answered defensibly, and what exactly blocks an answer?

SENSORS AND TELEMETRY

Proposed. Scope: all catalogue decisions and their required rooms, assets, services, meters and owner sources. Evidence: current inventory, mapping, health, cadence, calibration/functional-check, permission and provenance status for each required point. Timing: quality evaluated over each question's real decision window. Use only current mapped points with aligned clocks and current commissioning/calibration or functional checks. Report uptime, gaps, drift, relocation, maintenance and provenance. Exclude stale, failed, unmapped or indirect points; adjacent-space data is context only.

AUTHORITATIVE SUPPORTING DATA

Owners: asset/lifecycle, CMMS, BMS, meter/utility, room/service, incident/safety and data-governance owners as relevant. Records: explicit evidence requirement, source inventory, permission, retention, quality, conflict and owner status for every claim. Preserve source owner, effective/observed/retrieved time, permission, provenance, staleness and unresolved conflicts. Owner-issued status and competent decisions prevail. The May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation: calculation. Interval: current reporting and planning cycle. Minimum evidence: a question-to-evidence specification and current source availability. Method: audit each claim; classify supported, qualified, owner-deferred or not assessable; name the smallest remedy and whether a no-sensor source can answer it. State baseline or comparator, coverage, assumptions, conflicts and uncertainty. Keep observation, calculation/estimate, comparison/diagnosis, forecast and recommendation distinct. Safe failure: clarify, qualify, defer to the named owner or return not assessable; do not extrapolate beyond valid mapping or time coverage.

ANSWER BOUNDARY

Do not operate assets, acknowledge alarms, issue work, grant access, certify compliance or safety, identify people or infer individual behaviour. Do not fill gaps with general knowledge, model guesses or adjacent-room data, or present estimates as audited, compliant, safe or approved. Telemetry never authorises action; missing, stale, conflicting or unauthorised evidence requires a limited answer or owner deferral.

WHY THIS IS USEFUL

Facilities decision: prevent unsupported claims and create a prioritised evidence-remediation plan.

4. Catalogue summary and implementation requirements

The 80 questions form a facilities decision catalogue, not an automated control or authorisation system. Each answer must pass the same evidence, authority and safe-failure gates.

Asset status, requests and overview	FM-001 to FM-010	Prioritise review, validate evidence and coordinate owners.
Space use and service demand	FM-011 to FM-020	Prepare proportionate service checks from anonymous aggregate demand.
Environment and comfort	FM-021 to FM-030	Qualify room/zone evidence before inspection or intervention review.
Cleaning, waste and amenities	FM-031 to FM-040	Target service inspection without inferring hygiene or individual behaviour.
Maintenance	FM-041 to FM-050	Prepare competent inspection, work windows and handback evidence.
Contractors and handovers	FM-051 to FM-060	Check prerequisites and impact while permit/access owners retain authority.
Incidents and closures	FM-061 to FM-070	Coordinate facilities actions under certified and command sources.
Lifecycle and sustainability	FM-071 to FM-080	Support planning with measured service, condition, cost and uncertainty.

Minimum implementation gates

1. MAP

Current room, asset, meter, served-zone and dependency mapping.

2. TIME

Source owner, effective/observed/retrieved time and aligned clocks.

3. QUALITY

Commissioning/calibration or functional check, uptime, gaps, drift and maintenance state.

4. PERMISSION

Authorised access, privacy minimisation and restricted-source handling.

5. METHOD

Named operation, interval, minimum evidence, comparator and uncertainty.

6. FAILURE

Clarify, qualify, owner-defer or return not assessable when a gate fails.

5. Evidence, permissions and answerability rules

These rules apply to every record and define when the talking building may answer, must qualify, or must defer.

AUTHORITATIVE PRECEDENCE

Current owner-issued CMMS, asset, BMS, utility, room, service, permit, security, safety, incident and competent-professional sources govern their domains. Telemetry can support but never override them.

PROPOSED CAPABILITY

Sensors, integrations and analytics remain proposed until current deployment, mapping, commissioning, permission and health are independently verified.

SPATIAL AND TEMPORAL VALIDITY

Use exact asset, room, served-zone, meter and service-path scope. Never substitute corridor or adjacent-room evidence. Preserve clocks, effective/observed/retrieved times and decision-window completeness.

DATA QUALITY AND PROVENANCE

State calibration/functional checks, uptime, gaps, drift, relocation, maintenance, source owner and conflicts. Exclude stale, failed, clipped, unmapped or unauthorised points.

PRIVACY AND AUTHORITY

Use anonymous aggregate occupancy only where proportionate. Never identify or track people, infer individual behaviour, operate systems, authorise work/access, acknowledge alarms or certify compliance/safety.

SAFE FAILURE

When evidence or authority is missing, stale, conflicting or insufficient, ask the minimum clarification, provide a limited answer, defer to the named owner or return not assessable.

HISTORICAL PLAN LIMIT

The May 2021 Abacws floor plan is a historical spatial reference only. It cannot establish current room purpose, ownership, access, capacity, services, hazards, maintenance, closures or emergency status.

OUTPUT DISCIPLINE

Separate direct observation, authoritative lookup, calculation, estimate, forecast, comparison, diagnosis and recommendation. Show uncertainty, assumptions, comparators and the concrete facilities decision supported.